

Research Statement

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1 Introduction

Cyber-physical systems (CPS) are a class of systems that integrate software components with sensing and control to interact with the physical world. This includes isolated intelligent systems like autonomous vehicles (AVs), as well as large distributed systems like power grids, which must sense and react to faults in real time. The benefits of CPS coupled with the decreasing cost of computation and communication have led to their recent proliferation. However, software components that have the ability to effect changes in the physical world also have the potential to cause harm to the environment, humans, and their property. My research is focused on ensuring that we can continue to reap the benefits of such systems, while not exposing humans to unnecessary risks. In particular, I am focused on the integration of machine learning (ML) models with such systems, as their adoption is rapidly increasing thanks to recent advances in ML that enable more efficient and adaptable computation.

2 Future Directions

The functional performance of DNNs is highly dependent on environmental state, which has major implications for any safety-critical CPS. While there is a growing trend toward the use of end-to-end foundation models [1], such models do not have unlimited generalization capabilities [2] and even if they did, the power, memory, and execution time required may limit their use in practice. However, the converse approach of selecting from many small, domain-specialized models also has challenges. First, an ML model may be required for state estimation, which is not guaranteed to be accurate. Assuming an appropriate model is selected, less execution time may be available for downstream tasks in the processing pipeline. In the case where tasks are released at different rates, there is the question of how long state estimates are valid, and in complex systems with many tasks and expected environmental states, it may not be possible to develop individual models for each. My plan is to conduct a two-pronged approach to investigate these problems and maintain safety in ML-enabled CPS.

Dynamically scheduling DNNs An ensemble of specialized DNN models can be considered as a scheduling problem [3]. Consider the system in Fig. 1: solid lines represent precedence constraints, and dashed lines represent *logical dependencies* between components where the result of one ML model is statistically dependent on the result of another. Given the result of one of these networks, alternate models can be selected for execution, but the entire workload must be schedulable. Formulated this way, techniques developed for conditional directed acyclic graphs [4] could be modified to make solutions broadly applicable to a variety of DNN task sets. Two sub-problems I intend to investigate are minimizing the make span of a set of DNN tasks and minimizing power consumption, given a make span constraint.

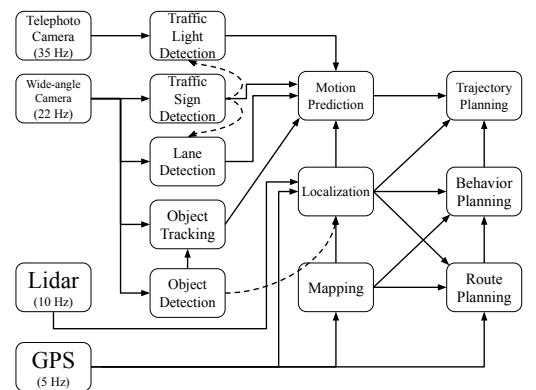


Figure 1: Example of an autonomous driving pipeline with logical dependencies as dashed lines [3].

Adaptable DNNs Previous works have proposed DNNs that can fuse data from many sources and generate outputs to satisfy multiple tasks in a single inference [5]. This has been used to help develop large end-to-end models; however, my proposal is to introduce training and inference-time modifications that will create a hybrid between monolithic models and pipelines of specialists. For example, gated convolutions [6] could be used to weight data from one or more sensors more heavily based on internal state estimates from a module internal to the DNN with its own unique loss function, trained simultaneously with the rest of the network. Additionally, as intermediate estimates become available at internal layers, the end-to-end model could terminate early if the state is favorable or switch to a different backbone by internalizing the decision logic currently used for early exits in anytime neural networks [7]. The ability to incorporate execution time estimation and nonfunctional constraints into an end-to-end DNN could allow for a monolithic network that can self-adapt based on rules learned from traditional scheduling techniques but still provide the advantages of a monolithic architecture.

Physical Test Bed It is important to show that the solutions identified by the previous two approaches are applicable to real-world CPS. I propose constructing an autonomous vehicle demonstrator that can verify the advantages conveyed by the aforementioned research directions. The novelty will be a focus on evaluating safety-critical edge cases, instead of general driving conditions. To accomplish this, I propose using a 1:10 to 1:16 scale vehicle and an environment capable of introducing challenging dynamics, e.g., icy conditions. Furthermore, the test bed should allow the evaluation of other aspects of intelligent transportation systems such as vehicle charging, battery health monitoring, and vehicle-to-vehicle and vehicle-to-infrastructure communication.

References

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